

Lecture 8

In Class Exercises



Exercise 1

- What are the minimum and maximum numbers of elements in a heap of height h ?

Exercise 2

- Show that an n -element heap has height $\lg n$.

Exercise 3

- Build a heap from the following array and argue if it is a max heap:
- {33, 19, 20, 15, 13, 10, 2, 13, 16, 12}

Exercise 4

- Suppose that the objects in a max-priority queue are just keys. Illustrate the operation of MAX-HEAP-INSERT($A, 10$) on the heap $A = \{15, 13, 9, 5, 12, 8, 7, 4, 0, 6, 2, 1\}$.